FDM Defect Detection – Team Work Plan (v2)

CSCI 6379 · Two-week sprint: Fri Oct 23 – Fri Nov 6, 2026 · Prepared by Hena Parween

1. Where we are

The project follows the proposal: classify six FDM defects (stringing, warping, layer shifting, under-extrusion, over-extrusion, bed adhesion) with transfer-learned MobileNet and ResNet (RQ1, RQ2), and map each predicted defect to corrective actions with a rule table that is checked against an independent reference (RQ3).

Item	Status
Real Roboflow data in the repo	Done. 6,132 images of the six classes from the 22,048-image export.
70/15/15 stratified split	Done, seed 42. Split by source photo: 103 photos sat in more than one Roboflow split, and now no photo appears in two of our splits.
Class weights	Done. Classes are imbalanced (bed adhesion 341 vs. under-extrusion 1,544 images), so the weights matter.
EDA figures, 18 scenarios, independent reference	Done (figures/, docs/).
Rule table (RQ3)	Not started. Template at data/rules/rule_table.json.
Models and test-set predictions (RQ1, RQ2)	Not started.

Ignore the earlier Oct 15 handoff note and its 100% agreement result. The data at that time was synthetic (computer-drawn shapes, because the real zip had not been unpacked), and the rules being scored were written by the same person as the reference. Both problems are fixed; the old files are kept in archive/synthetic_data/.

2. Who does what

Person	Background	Owns	RQ	Report sections
Hena	Computer science	Data pipeline, simulation, RQ3 agreement evaluation	RQ3 evaluation	Data; Experimental Setup; RQ3 results
Harrish	Computer science	MobileNet and ResNet training, metrics, inference script, rule-engine code	RQ1, RQ2	Methods; Results for RQ1 and RQ2
Agya	Manufacturing	Class audit, defect definitions, correction rule table, error interpretation	RQ3 rules	Background; Related Work; rule discussion

3. Tasks by person

Agya (manufacturing): defect knowledge and rules

- **Class audit (by Tue Oct 27).** Look at about 50 random images per class in data/processed/train/. Warping vs. bed adhesion and under- vs. over-extrusion are the likeliest mix-ups. Write problems to data/rules/class_audit.csv with the columns image_path, issue, suggested_class. Hena re-runs the cleaning with your findings.
- **Rule table (by Fri Oct 30).** Fill in data/rules/rule_table.json. For each defect give a primary action, up to two secondary actions, the direction (INCREASE, DECREASE, ENABLE or CHECK), a typical range and at least one citation (Jin et al. 2019, Brion et al. 2022, De La Rosa et al. 2024, or troubleshooting guides). Use the

parameter names in the file.
- **Independence rule:** do not open src/independent_reference.py or docs/INDEPENDENT_REFERENCE_TABLE.md until the rule table is submitted. RQ3 only means something if the two were written separately.
- **Error interpretation (Nov 4–5).** Explain the models' confusion matrices in physical terms.
- **Write:** Background, Related Work, and the rule-table discussion.

Harrish (computer science): models and code

- **Load data** from data/manifests/data_split_manifest.csv (the split column) or the data/processed/{train,val,test}/ folders. Use the class weights in data/manifests/class_distribution.json in the loss.
- **Train both models (by Fri Oct 30)** with identical settings: optimizer, learning rate, epochs, early stopping and augmentation. Train in two stages: first only the new head, then fine-tune deeper layers.
- **Evaluate (by Tue Nov 3):** accuracy, macro precision/recall/F1, per-class recall, confusion matrix, and average inference time per image on the same hardware for both models.
- **Predictions file (by Tue Nov 3):** run each model on the test split and write reports/predictions_<model>.csv with the columns image_path, true_class, pred_class, confidence (the softmax probability of the predicted class). Hena's simulation runs directly on these files.
- **Rule engine and inference script:** image → class + confidence → recommendation, reading the rules from data/rules/rule_table.json with the 65% confidence threshold.
- **Write:** Methods and the RQ1/RQ2 results.

Hena (computer science): data and simulation

- **Data (done).** Real data loaded, cleaned, split by source photo, class weights computed, EDA figures regenerated.
- **Apply the class audit (Oct 27–28)** and re-run the pipeline. Re-share the manifest if anything changes.
- **Scenario evaluation (Oct 30–31):** score the rule table against the reference on the 18 scenarios: primary-action agreement, any-action overlap and Cohen's kappa.
- **End-to-end simulation (Nov 3–4):** replay each model's test predictions through the rules and score every recommendation against the reference for the image's true class. Reports agreement, agreement when the system acts, and the abstention rate below 65% confidence, for MobileNet vs. ResNet.
- **Write (Nov 2–6):** Data, Experimental Setup and RQ3 results, with figures.

4. Two-week schedule

Dates	Hena	Harrish	Agya
Oct 23–27	Share this plan and the repo; data section draft	Training script; first ResNet run	Class audit → class_audit.csv
Oct 28–30	Apply audit, re-run split	Both models trained	Rule table complete
Nov 2–3	Scenario evaluation; RQ3 figures	Metrics, inference time, predictions CSVs	Background and Related Work draft
Nov 4–5	End-to-end simulation for both models	Rule engine and inference script; RQ1/RQ2 write-up	Confusion-matrix interpretation
Nov 6	Merge the report; final check	Repo cleanup and README	Rule discussion section

5. Handoff files

File	Written by	Used by	Due
data/manifests/data_split_manifest.csv	Hena	Harrish	Ready
data/rules/class_audit.csv	Agya	Hena	Oct 27
data/rules/rule_table.json	Agya	Harrish, Hena	Oct 30
reports/predictions_mobilenet.csv, reports/predictions_resnet.csv	Harrish	Hena	Nov 3
reports/AGREEMENT_RATE_EVALUATION.json	Hena	Everyone (report)	Nov 4

Reproduce the data and RQ3 steps from the repo root with: `python3 run_hena_pipeline.py` (needs the Roboflow zip in dataset/).